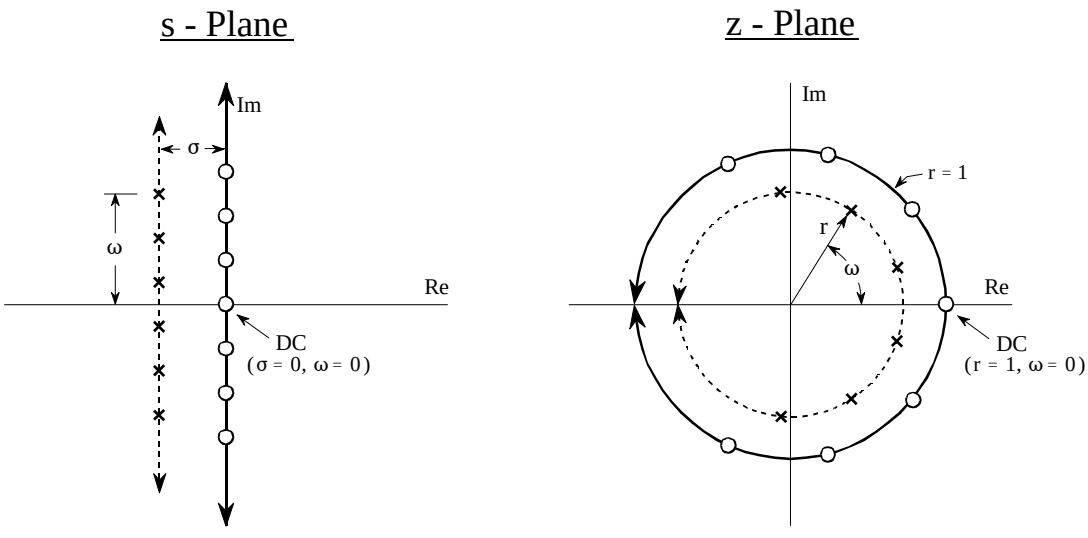

FIGURE 33-2
Relationship between the s-plane and the z-plane. The s-plane is a rectangular coordinate system with σ expressing the distance along the real (horizontal) axis, and ω the distance along the imaginary (vertical) axis. In comparison, the z-plane is in polar form, with *r* being the distance to the origin, and ω the angle measured to the positive horizontal axis. Vertical lines in the s-plane, such as illustrated by the example poles and zeros in this figure, correspond to circles in the z-plane.

unit circle (that is, $r = 1$). Vertical lines in the left half of the s-plane correspond to circles inside the z-plane's unit circle. Likewise, vertical lines in the right half of the s-plane match with circles on the outside of the z-plane's unit circle. In other words, the left and right sides of the s-plane correspond to the interior and the exterior of the unit circle, respectively. For instance, a continuous system is unstable when poles occupy the *right half* of the s-plane. In this same way, a discrete system is unstable when poles are *outside* the unit circle in the z-plane. When the time domain signal is completely real (the most common case), the upper and lower halves of the z-plane are mirror images of each other, just as with the s-domain.

Pay particular attention to how the frequency variable, ω, is used in the two transforms. A *continuous* sinusoid can have any frequency between DC and infinity. This means that the s-plane must allow ω to run from negative to positive infinity. In comparison, a *discrete* sinusoid can only have a frequency between DC and one-half the sampling rate. That is, the frequency must between 0 and 0.5 when expressed as a fraction of the sampling rate, or between 0 and π when expressed as a natural frequency (i.e., $\omega = 2\pi f$). This matches the geometry of the z-plane when we interpret ω to be an angle expressed in *radians*. That is, the positive frequencies correspond to angles of 0 to π radians, while the negative frequencies correspond to 0 to -π radians. Since the z-plane express frequency in a different way than the s-plane, some authors use different symbols to